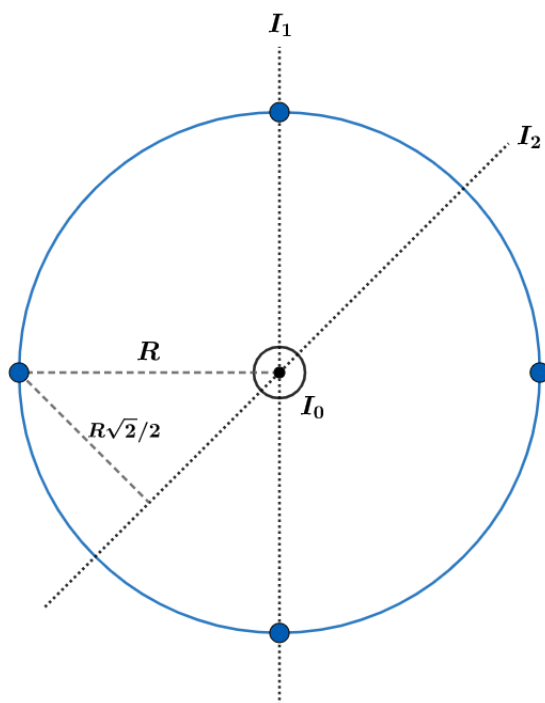


2022B F=ma Exam: Problem 2

Kevin S. Huang



From the definition of moment of inertia $I = \sum_i m_i r_i^2$,

$$I_0 = 4 \cdot m R^2$$

$$I_1 = 2 \cdot m R^2$$

$$I_2 = 4 \cdot m \left(\frac{R\sqrt{2}}{2} \right)^2 = 2mR^2$$

so $I_1 = I_2 = I_0/2$ and the answer is A.